

Understanding Linux Containers

Linux containers is an operating system level virtualisation method that is used to run multiple, isolated Linux systems on a control host with just a single Linux kernel.



If you look at the dictionary's definition of containers, it would be 'an object for holding and transporting something'. In the same way, Linux containers (mostly abbreviated as LXC in technical language) help in providing virtualisation to the Linux operating system. Tough to comprehend? Wait! Let's suppose that your enterprise requires to move your code from one environment to another, or that you want to keep your machines and servers tidy and clean. Whether these systems are for development or production purposes, or if you want to try out some different application or development stack on your regular machine without contaminating it, the solution is containers. Linux containers empower enterprises and companies in wrapping and isolating their products with all the necessary files used for running them. This makes it super easy to move containerised code between different environments and, at the same time, retain their full functionality. Technically, containers are created for the services that your server will provide, and the underlying hardware resources or kernel will be shared by all the containers.

Most of the time, Linux containers are mistaken for virtual machines or VMs. A virtual machine is just an emulation of a real computer that acts and responds like a real machine. The main difference between a container and

a VM is that containers share the host machine's kernel with all the other containers deployed on it, while VMs have many operating systems run on one machine with the help of a hypervisor. This feature of containers makes them light in weight compared to VMs and also helps in effective memory management. Containers have a number of benefits, some of which have been enumerated below:

1. Usually, containers require memory in megabytes, whereas a virtual machine can occupy several gigabytes. So, a single server can host a far greater number of containers, compared to how many VMs it can host.
2. Since containers are lightweight, the boot up time taken is also low. Further, they can be shut instantly. In comparison, virtual machines take a much longer time in booting up and shutting down.
3. Containers also provide modularity in development. Instead of running an entire application in a single container, it can be split into modules and run on different containers — for example, the front-end, the database, etc. This not only increases the performance of an application but also makes maintenance of the code easier.

Because of this, Linux containers are gaining significant ground in today's world, and a number of corporations are making use of containerisation technology